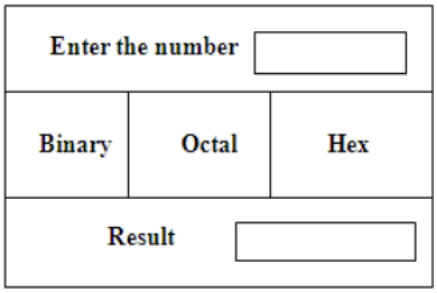
Experiment No. 11

Aim : Implementation of programs demonstrating different types of layout managers.

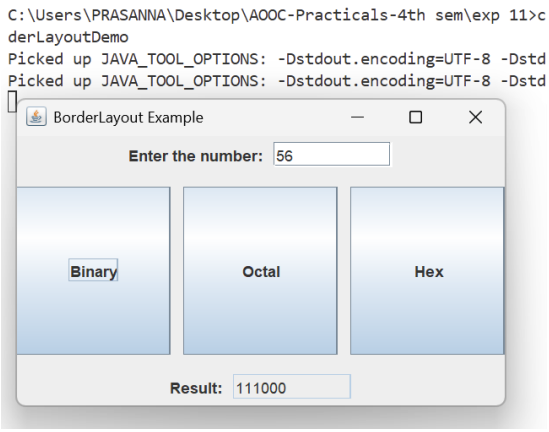
1. Border Layout: Implement following Example of Border Layout

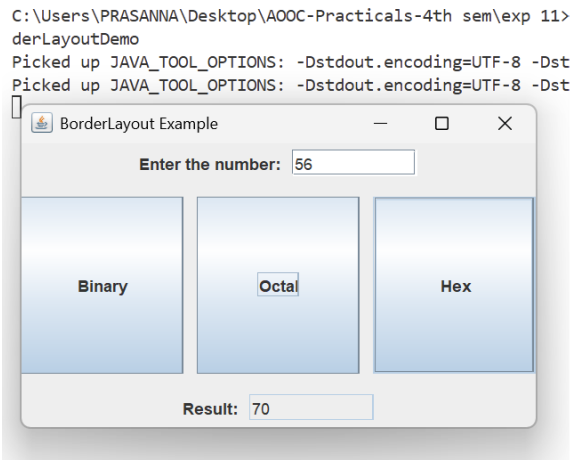


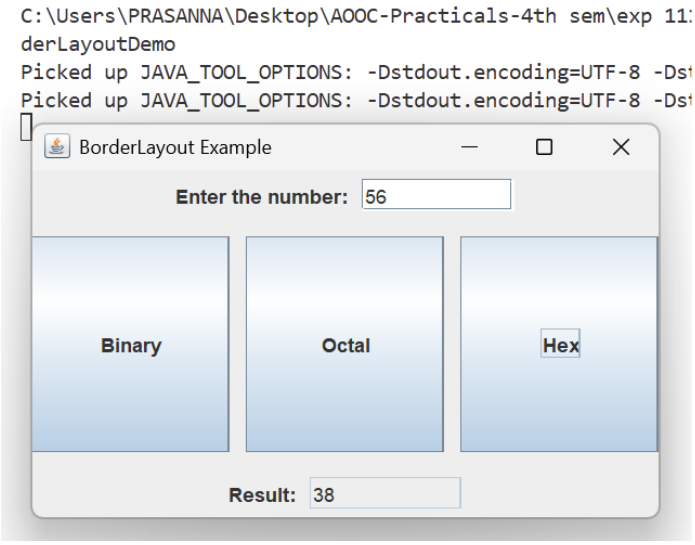

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class BorderLayoutDemo extends JFrame implements ActionListener {

    JTextField inputField, resultField;
    JButton binaryBtn, octalBtn, hexBtn;

    public BorderLayoutDemo() {

        setTitle("BorderLayout Example");
        setSize(400, 250);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        setLayout(new BorderLayout(10, 10));

        // Top Panel (NORTH)
        JPanel topPanel = new JPanel();
        topPanel.add(new JLabel("Enter the number: "));
        inputField = new JTextField(10);
        topPanel.add(inputField);
        add(topPanel, BorderLayout.NORTH);

        // Center Panel (CENTER)
        JPanel centerPanel = new JPanel(new GridLayout(1, 3, 10, 10));

        binaryBtn = new JButton("Binary");
        octalBtn = new JButton("Octal");
```

```
        hexBtn = new JButton("Hex");

        centerPanel.add(binaryBtn);
        centerPanel.add(octalBtn);
        centerPanel.add(hexBtn);

        add(centerPanel, BorderLayout.CENTER);

        // Bottom Panel (SOUTH)
        JPanel bottomPanel = new JPanel();
        bottomPanel.add(new JLabel("Result: "));
        resultField = new JTextField(10);
        resultField.setEditable(false);
        bottomPanel.add(resultField);

        add(bottomPanel, BorderLayout.SOUTH);

        // Add action listeners
        binaryBtn.addActionListener(this);
        octalBtn.addActionListener(this);
        hexBtn.addActionListener(this);

        setVisible(true);
    }

    @Override
    public void actionPerformed(ActionEvent e) {
        try {
            int num = Integer.parseInt(inputField.getText());

            if (e.getSource() == binaryBtn) {
                resultField.setText(Integer.toBinaryString(num));
            } else if (e.getSource() == octalBtn) {
                resultField.setText(Integer.toOctalString(num));
            } else if (e.getSource() == hexBtn) {
                resultField.setText(Integer.toHexString(num).toUpperCase());
            }

        } catch (NumberFormatException ex) {
            JOptionPane.showMessageDialog(this, "Please enter a valid number!");
        }
    }

    public static void main(String[] args) {
        new BorderLayoutDemo();
    }
```

}


C:\Users\PRASANNA\Desktop\AOOC-Practicals-4th sem\exp 11>c
derLayoutDemo
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstd
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstd
BorderLayout Example
Enter the number: 56
Binary
Octal
Hex
Result: 111000

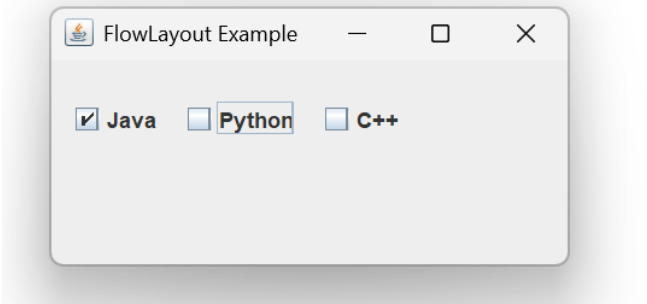

C:\Users\PRASANNA\Desktop\AOOC-Practicals-4th sem\exp 11>
derLayoutDemo
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dst
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dst
BorderLayout Example
Enter the number: 56
Binary
Octal
Hex
Result: 70


C:\Users\PRASANNA\Desktop\AOOC-Practicals-4th sem\exp 11:
derLayoutDemo
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dst
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dst
BorderLayout Example
Enter the number: 56
Binary
Octal
Hex
Result: 38

2. FlowLayout: Create a Java program using FlowLayout (aligned left, with horizontal gap 10px and vertical gap 20px) that adds three checkboxes labeled "Java", "Python", and "C++" into the frame.

```
import javax.swing.*;
import java.awt.*;

public class FlowLayoutDemo extends JFrame {

    public FlowLayoutDemo() {

        setTitle("FlowLayout Example");
        setSize(300, 150);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        // FlowLayout: LEFT alignment, hgap = 10, vgap = 20
        setLayout(new FlowLayout(FlowLayout.LEFT, 10, 20));

        // Create Checkboxes
        JCheckBox java = new JCheckBox("Java");
        JCheckBox python = new JCheckBox("Python");
        JCheckBox cpp = new JCheckBox("C++");

        // Add to frame
        add(java);
        add(python);
        add(cpp);

        setVisible(true);
    }

    public static void main(String[] args) {
        new FlowLayoutDemo();
    }
}
```

```
C:\Users\PRASANNA\Desktop\AOOC-Practicals-4th s
Messages -cp "C:\Users\PRASANNA\AppData\Roaming
 "
```



3. GridLayout: Create a program that demonstrates the use of GridLayout. Display a 2x3 grid with the following numbers inside each box. Also, when the user clicks on any box, the number inside that box should swap with the number

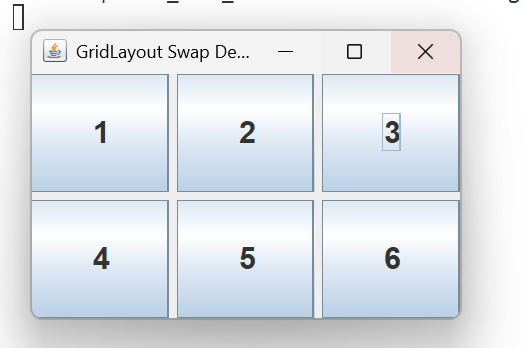
1	2	3
4	5	6

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class GridLayoutSwapDemo extends JFrame implements ActionListener {

    JButton[] buttons = new JButton[6];
    int firstIndex = -1; // to store first clicked button

    public GridLayoutSwapDemo() {

        setTitle("GridLayout Swap Demo");
        setSize(300, 200);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        // 2 rows, 3 columns
        setLayout(new GridLayout(2, 3, 5, 5));

        // Numbers to display
        for (int i = 0; i < 6; i++) {
            buttons[i] = new JButton(String.valueOf(i + 1));
            buttons[i].setFont(new Font("Arial", Font.BOLD, 20));
            buttons[i].addActionListener(this);
            add(buttons[i]);
        }

        setVisible(true);
    }

    @Override
    public void actionPerformed(ActionEvent e) {

        // Find which button was clicked
        for (int i = 0; i < 6; i++) {
            if (e.getSource() == buttons[i]) {
```

```
            if (firstIndex == -1) {
                // First click
                firstIndex = i;
                buttons[i].setBackground(Color.LIGHT_GRAY); // highlight
            } else {
                // Second click → swap
                String temp = buttons[firstIndex].getText();
                buttons[firstIndex].setText(buttons[i].getText());
                buttons[i].setText(temp);

                // Reset color
                buttons[firstIndex].setBackground(null);
                firstIndex = -1;
            }
            break;
        }
    }
  }

  public static void main(String[] args) {
      new GridLayoutSwapDemo();
  }
}
```

OUTPUT :

```
c:\Users\PRASANNA\Desktop\AOOC-Practicals-4th s
ridLayoutSwapDemo
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=
```




```
c:\Users\PRASANNA\Desktop\AOOC-Practicals-4th sem
ridLayoutSwapDemo
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTI
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTI
```

4. Write a GUI program to find the factorial of a given number using applet. (You will need
Java 8 to run applet)
How to compile and execute applet (You will need Java 8 to run applet)
Lab Manual: AOOC
Lab manual: Advanced Object-Oriented Concepts (Java), SY CSE, Sem IV, 2025-26, Computer Science & Engineering Dept., D.Y. Patil College of Engineering & Technology, Kasaba Bawada, Kolhapur
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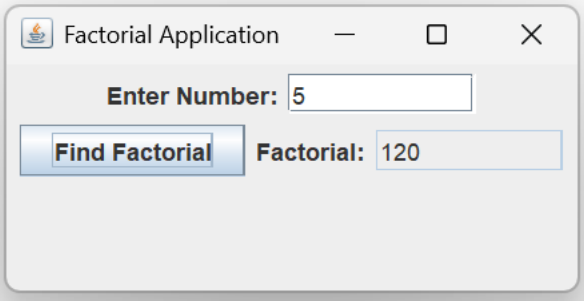
○ Compile the Applet - javac factorialappl.java
○ Run with Applet Viewer - appletviewer factorialappl.java
○ OR if using the HTML file -
■ Create an HTML File (if not using appletviewer)
■ Save below as factorial.html
<html>
<body>
<applet code="factorialappl.class" width="400" height="400"></applet>
</body>
</html>
■ appletviewer factorial.html

```
import java.applet.Applet;
import java.awt.*;
import java.awt.event.*;

public class factorialappl extends Applet implements ActionListener {

    Label l1, l2;
    TextField t1, t2;
    Button b1;

    public void init() {

        l1 = new Label("Enter Number:");
        l2 = new Label("Factorial:");

        t1 = new TextField(10);
        t2 = new TextField(10);
        t2.setEditable(false);

        b1 = new Button("Find Factorial");
```

```
        add(l1);
        add(t1);
        add(b1);
        add(l2);
        add(t2);

        b1.addActionListener(this);
    }

    public void actionPerformed(ActionEvent e) {
        try {
            int num = Integer.parseInt(t1.getText());
            long fact = 1;

            for (int i = 1; i <= num; i++) {
                fact *= i;
            }

            t2.setText(String.valueOf(fact));

        } catch (NumberFormatException ex) {
            t2.setText("Invalid Input");
        }
    }
}
```

OUTPUT :

```
c:\Users\PRASANNA\Desktop\AOOC-Practicals-4th sem
ialappl
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTI
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTI
```